

The Constants of Nature

$$\begin{array}{l}
d_{ca} = \frac{E_1}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \mathcal{K}_\theta^2 \right) \qquad A_{\text{modd}} = \mathcal{K}_1 \left(\frac{l_p^2 q_p m_p}{l_p m_p} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{K}_e^2 \right) \qquad E_p = \frac{eV}{\text{joule}} \left(\frac{t_p^2 q_p}{t_p^2} \right) \left(1 - \frac{1}{2} Re \left(i^{l^{tr}} \right) \right) \mathcal{K}_3 \mathcal{K}_4 \mathcal{K}_5 \qquad g_{de} = \frac{2\pi}{8K} \left(1 - 8 \left(\frac{2\pi}{W_{We}} \right) \mathcal{K}_1 \mathcal{K}_3 \mathcal{K}_4 \mathcal{K}_5 \right) \qquad c = \left(\frac{l_p}{t_p} \right) \left(1 + \sqrt{G_{G1}} (\mathcal{K}_1 - \mathcal{K}_2) \right) \qquad F_{90} = \text{farad} \left(1 + \frac{C_d}{6} (\mathcal{K}_1 + \mathcal{K}_3 + \mathcal{K}_4) \right) \qquad K_f = \frac{\mathcal{K}_1}{\pi} \left(\frac{t_p q_p}{l_p^2 m_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathcal{K}_1^2 + \mathcal{K}_2^2) \right) \qquad b' = \frac{1}{2\pi} \left(\frac{V_{peak}}{t_p T_p} \right) \left(1 + 18 \left(\frac{2\pi}{4s} \right)^2 (\mathcal{K}_1^2 - \mathcal{K}_3^2 - \mathcal{K}_4^2) \right) \\
eV : Hz = \frac{eV}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \mathcal{K}_\theta^2 \right) \qquad \mu_B = \frac{\mathcal{K}_1}{2} \left(\frac{t_p^2 q_p m_p}{l_p m_p} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{K}_e^2 \right) \qquad R_K = \frac{2\pi}{\mathcal{K}_1^2} \left(\frac{l_p^2 m_p}{t_p q_p^2} \right) \left(1 + \frac{1}{2} Re \left(i^{l^{tr}} \right) \right) \mathcal{K}_3 \mathcal{K}_4 \mathcal{K}_5 \qquad \frac{\mu_{de}}{\mu_N} = \frac{2\pi}{8K} \left(1 - 8 \left(\frac{2\pi}{W_{We}} \right) \mathcal{K}_1 \mathcal{K}_3 \mathcal{K}_4 \mathcal{K}_5 \right) \qquad \frac{1}{m} : Hz = \left(\frac{l_p}{t_p m} \right) \left(1 + \sqrt{G_{G1}} (\mathcal{K}_1 - \mathcal{K}_2) \right) \qquad H_{90} = \text{henry} \left(1 - \frac{C_d}{6} (\mathcal{K}_1 + \mathcal{K}_3 + \mathcal{K}_4) \right) \qquad \mu_{90} = \text{ohm} \left(1 - \frac{C_d}{6} (\mathcal{K}_1 + \mathcal{K}_3 + \mathcal{K}_4) \right) \qquad m_2 = (m_n - m_1) \left(1 - 35 \frac{C_d}{6} (\mathcal{K}_1 + \mathcal{K}_3 + \mathcal{K}_4) \right) \qquad \frac{e}{h} = \mathcal{K}_1 \left(\frac{t_p^2 q_p}{l_p^2 m_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathcal{K}_1^2 + \mathcal{K}_2^2) \right) \qquad e_{\bar{h}} = \mathcal{K}_1 \left(\frac{t_p^2 q_p}{l_p^2 m_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathcal{K}_1^2 + \mathcal{K}_2^2) \right) \qquad \Phi_0 = \frac{\pi}{\mathcal{K}_1} \left(\frac{l_p^2 m_p}{t_p q_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathcal{K}_1^2 + \mathcal{K}_2^2) \right) \qquad \Phi_0 = \frac{\pi}{\mathcal{K}_1} \left(\frac{l_p^2 m_p}{t_p q_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathcal{K}_1^2 + \mathcal{K}_2^2) \right) \qquad \tilde{h} = \frac{1}{\mathcal{K}_1} \left(\frac{l_p^2 C m_p}{t_p q_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathcal{K}_1^2 + \mathcal{K}_2^2) \right) \qquad \tilde{h} = \frac{1}{\mathcal{K}_1} \left(\frac{l_p^2 C m_p}{t_p q_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathcal{K}_1^2 + \mathcal{K}_2^2) \right) \qquad \dot{h} = \frac{2\pi}{\mathcal{K}_1} \left(\frac{l_p^2 C m_p}{t_p q_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathcal{K}_1^2 + \mathcal{K}_2^2) \right) \qquad \dot{h} = \frac{2\pi}{\mathcal{K}_1} \left(\frac{l_p^2 C m_p}{t_p q_p} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathcal{K}_1^2 + \mathcal{K}_2^2) \right) \qquad b = 2\pi \left(\frac{l_p T_p}{t_{peak}} \right) \left(1 + P^2 (\mathcal{K}_1^2 + \mathcal{K}_3^2 + \mathcal{K}_4^2) \right) \qquad c_2 = 2\pi \left(\frac{l_p T_p}{t_{peak}} \right) \left(1 + P^2 (\mathcal{K}_1^2 + \mathcal{K}_3^2 + \mathcal{K}_4^2) \right) \qquad \frac{1}{m} : K = 2\pi \left(\frac{l_p T_p}{t_p} \right) \left(1 + P^2 (\mathcal{K}_1^2 + \mathcal{K}_3^2 + \mathcal{K}_4^2) \right) \qquad K : \frac{1}{m} = \frac{1}{2\pi} \left(\frac{K}{t_p T_p} \right) \left(1 - P^2 (\mathcal{K}_1^2 + \mathcal{K}_3^2 + \mathcal{K}_4^2) \right) \qquad k_B = \frac{1}{2\pi} \left(\frac{1}{t_p T_p} \right) \left(1 - P^2 (\mathcal{K}_1^2 + \mathcal{K}_3^2 + \mathcal{K}_4^2) \right) \\
Hz : eV = 2\pi \frac{eV}{\text{joule}} \left(\frac{t_p^2}{s t_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_\theta^2 \right) \qquad \frac{\mu_{tr1}}{\mu_N} = S \left(D_{D0} + \pi \right) \left(\frac{m_+}{m_{tr1}} \right) \left(1 + \frac{4}{L_{L1}} \left(\frac{32}{4\pi} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_N} = \mathcal{K}_1 \mathcal{K}_2^2 \left(2D_{D0} + 2\pi \right) \left(\frac{m_+}{m_{tr1}} \right) \left(1 - \frac{18}{3} \left(\frac{4\pi}{35} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_B} = 2 \mathcal{K}_1 \mathcal{K}_2^2 \left(2D_{D0} + 2\pi \right) \left(\frac{m_+}{m_{tr1}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_B} = \mathcal{K}_1 \mathcal{K}_2^2 \left(4D_{D0} + 4\pi \right) \left(\frac{m_+}{m_{tr1}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_N} = -\frac{S}{C_{CFP}} \left(\frac{m_+}{m_n} \right) \left(1 + 15 \left(\frac{2\pi}{8K} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_N} = -\frac{C_{CFP}}{S} \left(\frac{m_n}{m_+} \right) \left(1 + 15 \left(\frac{2\pi}{8K} \right) \mathcal{K}_e^2 \right)^{-1} \qquad \frac{\mu_{tr1}}{\mu_N} = \frac{C_{CFP}}{P_{up}} \left(\frac{m_n}{m_+} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{P_{up}}{C_{CFP}} \left(\frac{m_+}{m_n} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right)^{-1} \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{C_{CFP}}{P_{up}} \left(\frac{m_n}{m_+} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{P_{up}}{C_{CFP}} \left(\frac{m_+}{m_n} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right)^{-1} \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{C_{CFP}}{P_{up}} \left(\frac{m_n}{m_+} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{P_{up}}{C_{CFP}} \left(\frac{m_+}{m_n} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right)^{-1} \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{C_{CFP}}{P_{up}} \left(\frac{m_n}{m_+} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{P_{up}}{C_{CFP}} \left(\frac{m_+}{m_n} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right)^{-1} \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{C_{CFP}}{P_{up}} \left(\frac{m_n}{m_+} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{P_{up}}{C_{CFP}} \left(\frac{m_+}{m_n} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right)^{-1} \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{C_{CFP}}{P_{up}} \left(\frac{m_n}{m_+} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{P_{up}}{C_{CFP}} \left(\frac{m_+}{m_n} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right)^{-1} \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{C_{CFP}}{P_{up}} \left(\frac{m_n}{m_+} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right) \qquad \frac{\mu_{tr1}}{\mu_e} = \frac{P_{up}}{C_{CFP}} \left(\frac{m_+}{m$$

legend

$m_p = 5.39125836832313 \times 10^{-34}$ s	Planck time
$l_p = 1.61625918175645 \times 10^{-35}$ m	Planck length
$q_p = 1.87554596713962 \times 10^{-18}$ C	Planck charge
$T_p = 1.41678698590795 \times 10^{32}$ K	Planck temperature
$m_p = 2.17642683817579 \times 10^{-8}$ kg	Planck mass
$\boxtimes = 0.919999199973626 \times 10^{-7}$	hyperbolic inversion boundary
$\boxtimes$	number of dimensions
$\boxdot$	break in scale symmetry
$\chi_1 = 0.0854245431533304 \dots$	1 st hyperbolic partition constant
$\chi_2 = 3.66756753485501 \dots$	2 nd hyperbolic partition constant
$\chi_3 = 1.87649603900417 \dots + 0.46615262615972 \dots i$	3 rd hpc
$\chi_4 = 1.87649603900417 \dots - 0.46615262615972 \dots i$	4 th hpc
$\chi_5 = 4.47826244916751 \dots$	hyperbolic radius constant
$\chi_6 = 2.00316562310924 \dots$	hyperbolic radian constant
$G_{00} = 1.01494160640965 \dots$	Gieseking's constant
$V_6 = 0.29598321281930 \dots$	figure eight knot hyperbolic volume
$K = 0.0159655941772190 \dots$	Catalan's constant
$C_d = 0.291560904030818 \dots$	domino tiling constant
$D_d = 1.79162281206959 \dots$	dimer constant
$\omega_1 = 0.764974078528596 \dots + 1.32497062714087 \dots i$	omega_1 constant
$\omega_2 = 1.529954937057192 \dots$	omega_2 constant
$i^{i^{i^{\dots}}}$	power tower of i
$i = (-1)^{1/3}$	imaginary unit
$\varphi_1 = (-1)^{1/3}$	imaginary golden ratio
$\varphi = 1.61803398874989 \dots$	golden ratio
$\varphi = 1.9145926358979 \dots$	Archimedes' constant
$R_{up} = 2.29558714939263 \dots$	universal parabolic constant
$S = 5.24411510858423 \dots$	arc length of the unit lemniscate
$L = 2.62025755429219 \dots$	lemniscate constant
$L_1 = 1.31102877714605 \dots$	1 st lemniscate constant
$L_2 = 0.599070117367796 \dots$	2 nd lemniscate constant
$C_0 = 0.847213084793979 \dots$	ubiquitous constant
$G_{00} = 0.83462684167403 \dots$	Gauss's constant
$W_{We} = 0.47949379987920 \dots$	Weierstrass constant
$C_{R1} = 0.9981360445989509 \dots$	Ramanujan's 1 st continued fraction constant
$\epsilon_L = 0.110000000000000 \dots$	Liouville's constant
$M = -1.74756459463318 \dots$	Madelung constant
$\mathcal{P} = 1.22674201072035 \dots$	prime constant
$e = 2.71828182845904 \dots$	Euler's number
$x_n = 2.29316628741186 \dots$	1 st Foias constant
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$S = 0.822825249679847 \dots$	Sierpiński's constant
$D_{10} = 0.739085133215160 \dots$	Dottie number
$\epsilon_L = 0.66274341394391 \dots$	Laplace limit
$C_{RPP} = 1.199676802527577 \dots$	real fixed point of the hyperbolic cotangent
$J_{0,1} = 2.404825557695772 \dots$	1 st root of the Bessel function
$\alpha_F = 2.50290787509589 \dots$	alpha Feigenbaum constant
$\alpha_F = 0.66920160910299 \dots$	delta Feigenbaum constant
$\delta = 3.24697960371714 \dots$	silver constant
$P = 1.32471795724474 \dots$	plastic constant
$K_0 = 0.605443657196732 \dots$	QRS constant
$K_{-6} = 1.15655237442151 \dots$	Khinchin mean of order −6
$T_0 = 2.73150000000000 \times 10^{12}$ kelvin	freezing point of water
$p_0 = 1.00000000000000 \times 10^5$ pascal	1 bar
$p_1 = 1.01325000000000 \times 10^5$ pascal	standard atmospheric pressure
$E_1 = 6.09110229711386 \times 10^{-24}$ joules	energy associated with Δ_{CS}
$f_1 = 5.40000000000000 \times 10^{-14}$ Hz	frequency of max light conversion
$\lambda_{\text{peak}} = 4.96511423174427 \dots$	peak 2 of spectral radiance/unit wavelength
$\nu_{\text{peak}} = 2.82143937215207 \dots$	peak emission of spectral flux/unit frequency
$n!$ = factorial function	
$\ln$ = derangement function	
$\log(x)$ = natural logarithm	$\text{lm} = \text{lumen} = \text{cd} \cdot \text{sr}$
$\Gamma(x)$ = gamma function	$N = \text{newton} = \text{m kg/s}^2$
$\zeta(x)$ = Riemann zeta function	$\mathcal{N} = \text{noether} = \text{m kg/s}$
	$\Omega = \text{ohm} = \text{m}^2 \text{ kg/s}^2 \text{ C}^2$
units	$\text{Pa} = \text{pascal} = \text{kg/s}^2 \text{ m}^2$
s = second	$S = \text{siemens} = \text{s C}^2/\text{m}^2 \text{ kg}$
m = meter	$T = \text{tesla} = \text{kg/s}^2 \text{ C}$
C = coulomb	$V = \text{volt} = \text{m}^2 \text{ kg/s}^2 \text{ C}$
k = kelvin	$W = \text{watt} = \text{m}^2 \text{ kg/s}^3$
kg = kilogram	$\text{Wb} = \text{weber} = \text{m}^2 \text{ kg/s C}$
derived units	
MHz = megahertz = 1.00000000000000 ... $\times 10^6$ 1/s	
fm = femtometer = 1.00000000000000 ... $\times 10^{-15}$ m	
eV = electron-volt = 1.00000000000000 ... eV	
MeV = megaelectron-volt = 1.00000000000000 ... $\times 10^6$ eV	
GeV = gigaelectron-volt = 1.00000000000000 ... $\times 10^9$ eV	
$\frac{r_n}{r_1} = \left(\frac{4\pi}{\Gamma(5)}\right)^2$	$\frac{r_n}{r_1} = K$
$\frac{r_n}{r_1} = K$	$\frac{r_n}{r_1} = \sqrt{\frac{C_1}{C_2}}$
	$\frac{r_e}{r_{\text{der}}} = \sqrt{\frac{36}{2\pi}}$
$e_p = \frac{l_p^2}{t_p^2}$	
transform space	Physics Monastery
$V_{24} = \left(\frac{4\pi}{5!}\right) \left(\frac{4\pi}{15}\right) \left(\frac{4\pi}{35}\right) \left(\frac{4\pi}{18}\right) \left(\frac{4\pi}{3$	